LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Over voltage protection DC generating systems, over voltage protection AC generating systems, under voltage protection, under and over excitation protection, under and over frequency protection, differential current protection
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Over voltage protection DC generates systems, over voltage protection AC generating systems, under voltage protection, under and over excitation protection, under and over frequency protection, differential current protection.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of generator protection schematics; Safety: Do not touch live terminals; Instructor supervision: Monitor student diagram tracing.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a generator protection system?
(b) Name one device used for over-voltage protection in DC systems.
(c) Why is under-excitation protection necessary for synchronous generators?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Over voltage protection DC generating systems • Over voltage protection AC generating systems • Under voltage protection mechanisms • Under and over excitation protection • Under and over frequency protection	L & Ds	PPT, IB	15 min
(b)	• Differential current protection principles • Circuit protection device operation • Generator actuator protection systems • Engine starting system protection • Integration of protection relays	L & Ds	PPT, PDF	20 min
(c)	• Fault isolation procedures • Protection relay settings • System response to excitation faults • Frequency deviation handling • Review of protection hierarchy	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how differential current protection detects internal generator faults.
(b) Describe the sequence of events during an over-voltage condition in an AC generator.
(c) What are the consequences of ignoring under-excitation protection?

12. Home Assignment and Follow Up	Prepare a brief note on the specific protection requirements for the engine starting system as covered in the next lesson plan sequence.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Motor actuators: - Internal actuators, rotary actuators, actuator gearing, limit s/w, breaks, clutches.	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min